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SANTHOSH BELL

Junior Robotics Engineer

TECHNICAL SKILLS

Programming: C++, Python, Embedded C
Embedded & IoT: Arduino, ESP32/ESP8266, Raspberry Pi
Protocols: UART, I2C, SPI, MQTT, HTTP
Robotics: ROS (Basics), Gazebo, Sensor & Motor Integration
Tools: Arduino IDE, EasyEDA, Proteus, Tinkercad
AI Basics: OpenCV (Basic), ML Concepts for Robotics

HARDWARE EXPERIENCE

- Sensor Integration (DHT11, Ultrasonic, IR, LDR, Gas Sensors)
- Motor Drivers & Power Systems
- PCB Design & Soldering
- Serial Communication (UART, I2C, SPI)

SOFT SKILLS

- Problem Solving
- Analytical Thinking
- Team Collaboration
- Project Documentation
- Technical Presentation

SUMMARY

Robotics & IoT enthusiast with hands-on experience in Arduino, ESP32, Raspberry Pi, and ROS. Strong foundation in Embedded Systems, IoT communication, and AI-based robotics concepts. Seeking a Junior Robotics Engineer position to apply practical knowledge in robot development, automation systems, and intelligent robotic solutions in real-world applications.

Education

- 2020-2021

Higher Secondary Certificate
Walker Higher Secondary SCHOOL

- 2021-2025

B.E Robotics And Automation
Loyola Institute of Technology And Science

PROJECTS

Smart Home Automation (IoT)

- ESP32-based appliance control with web dashboard
- MQTT communication & cloud integration

IoT Weather Monitoring System

- Sensor data collection & real-time cloud logging

ROS-Based Robot Simulation

- Publisher/Subscriber model implementation
- Robot simulation using Gazebo

Capstone Project – IoT + Robotics Integration

- Arduino + ESP + Raspberry Pi + ROS
- End-to-end working prototype with dashboard

LANGUAGES

Tamil
English